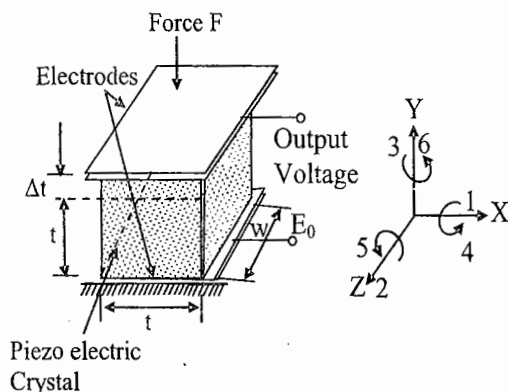




Piezo Electric Transducers

Piezo-electric transducers are active transducers. They work on the principle of piezo-electric effect. A piezo-electric material is one in which an electric potential appears across certain surfaces of a crystal if the dimensions of the crystal are changed by the application of a mechanical force.

- Each piezo-electric crystal has three sets of axes :
 - An optical axes
 - Three electrical axes (X) at 120° with each other and
 - Three mechanical axes (Y) at 120° with each other and at right angle to the respective electric axes.
- If an electric stress is applied in the direction of a electrical axis (X-axis), a mechanical strain is produced in the direction of the mechanical axis (Y), which is perpendicular to the relevant X-axis, conversely if a mechanical stress is applied along a Y-axis, electric charges appear on the faces of the crystal perpendicular to the relevant Y-axis.
- This effect is known as piezo-electric effect.



- The materials that exhibit a significant and useful piezo-electric effect are divided into two categories:
 - (i) Natural group

(ii) Synthetic group

- **Natural Group:**

Quartz and Rochelle salt

- **Synthetic Group :**

Lithium sulphate, ethylene diamine tartarate, Ammonium dihydrogen phosphate, dipotassium tartarate, potassium dihydrogen phosphate.

- Synthetic or artificial crystals, they themselves do not possess the piezo-electric property. By applying high voltages across the electrodes heating to higher temperature and slow cooling induces the property.
- With synthetic crystals, if the temperature is more than a threshold value called curie-point, the property is lost.
- Properties of crystal depends on
 - cutting of the crystal
 - Thickness of the crystal
 - Dimensions of the plate and
 - Means of mounting
- The piezo-electric effect can be made to respond to mechanical deformations of the material in many different modes.
- The modes can be:
 - Thickness expansion
 - Transverse expansion
 - Thickness shear and
 - Face shear
- The mode of motion effected depends on the shape of the body relative to the crystal axis and location of the electrodes.
- Mechanical deformation generates a charge and this charge appears as a voltage across the electrode.

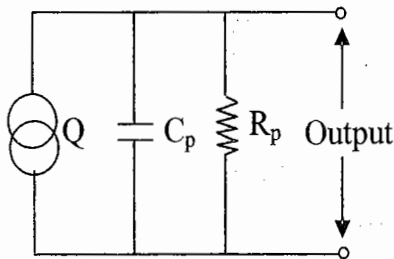


The voltage is $E = Q / C$

- The piezo-electric effect is direction sensitive. A tensile force produces a voltage of one polarity while a compressive force produces a voltage of opposite polarity.
- The magnitude and polarity of the induced surface charges are proportional to the magnitude and direction of the applied force F . The polarity of induced charges depends upon the direction of applied force.

Equivalent circuit of piezo-electric transducers:

The basic equivalent circuit of a piezo-electric transducer is shown in the following figure

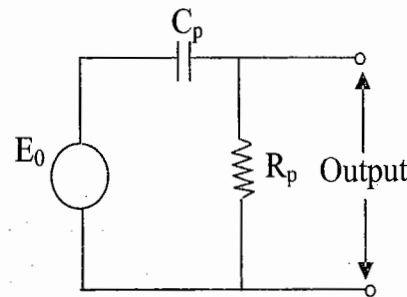


- The source is charge generator.
- The value of the charge is $Q = dF$
- The charge generated is across the capacitance C_p of the crystal and its leakage resistance R_p . The charge generator can be

replaced by an equivalent voltage source having a voltage of $E_o = \frac{Q}{C_p} = \frac{dF}{C_p}$

in series with a capacitance, C_p and resistance R_p .

The circuit is shown in the following figure.



Properties of Piezo-electric crystal

- Stability
- High output
- Insensitivity to temperature variations
- Insensitivity to variations in humidity
- Ability to be formed into a useable component

Most commonly used piezo-electric crystals:

- Quartz is the most stable piezo-electric material. However, its output is quite small.
- Rochelle salt provides highest output but it can be warm over limited temperature and humidity range.